Frequency-Guided Mutual Distillation for Lightweight Cross-Modal Remote Sensing Image Retrieval

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***Abstract*—Cross-modal remote sensing image retrieval (CMRSIR) is vital for Earth observation but remains challenged by the fundamental heterogeneity gap. Existing hashing methods often struggle to reconcile fine-grained representation with semantic consistency, primarily due to resolution-induced physical degradation and spatial–spectral discrepancies. To address these issues, we propose the frequency-guided mutual distillation (FG-MD) network, a lightweight framework designed for efficient and consistent feature alignment. The proposed approach first introduces a fine-grained refinement stage featuring a frequency-domain full-spectrum perceiver (FDFSP) module and an adaptive dynamic filter (ADF), which synergistically enhance high-frequency structural details and suppress environmental clutter. Subsequently, a dual-attention mutual distillation (DAMD) strategy is devised to enforce spatial-semantic consistency via symmetric collaborative learning without incurring additional inference overhead. Extensive experiments conducted across multiple benchmarks demonstrate that FG-MD consistently achieves state-of-the-art performance with only 22.05-M parameters, offering an optimal efficiency-accuracy tradeoff across all evaluated datasets. Furthermore, to validate generalization, we construct the indoor geometric structure (IGS) benchmark for GPS-denied UAV reconnaissance. This dataset constitutes a challenging isomorphic extension of remote sensing tasks, where FG-MD significantly outperforms competitive baselines under severe viewpoint variations.**

***Index Terms*—Cross-modal retrieval, frequency-domain learning, lightweight deep hashing, mutual distillation, remote sensing.**

I. Introduction

CROSS-MODAL image retrieval (CMIR) aims to transcend the perceptual limitations of single sensors by using a query from one modality to precisely retrieve semantically consistent samples from another [1], [2], [3], [4]. The fundamental challenge lies in overcoming the "heterogeneity gap" [5], [6], [7], [8], [9] caused by distinct imaging mechanisms, which manifests as severe shifts in low-level visual statistics and inconsistencies in high-level semantic alignment. By exploiting complementary information across modalities, CMIR plays an indispensable role in multimedia analysis and complex Earth observation tasks [5], [10], [11], [12], [13], [14], [15], [16].

In cross-modal remote sensing image retrieval (CMRSIR), a typical application scenario involves utilizing high-spatial-resolution panchromatic (PAN) images as queries to retrieve multispectral (MS) images rich in spectral information [5], [6], [7], [17]. Early metric learning approaches, such as the interpretation-fusion Siamese network (IFSN) [7], achieved high retrieval precision via high-dimensional real-valued features. However, their high computational and storage costs are ill-suited for real-time processing of massive datasets. Consequently, hashing-based paradigms [18], [19], [20], [21], [22] have become mainstream due to their compact binary codes and efficient Hamming distance calculations. Nevertheless, existing hashing methods struggle to reconcile fine-grained feature representation with cross-modal semantic consistency when confronting the significant resolution disparities and modality gaps between PAN and MS images, thereby limiting further performance gains [6], [23], [24], [25].

Specifically, two fundamental obstacles hinder the achievement of cross-modal consistency. We take the PAN-MS matching modality pair as an example for analysis. On the one hand, a profound modality gap and fine-grained semantic inconsistency persist between heterogeneous multimodal views [7], [17]. Constrained by physical sensor limitations, low-resolution MS images suffer from the irreversible degradation of high-frequency components (e.g., edges and shapes of small targets) during downsampling. This inherent information asymmetry prevents the extraction of delicate, fine-grained representations comparable to high-resolution PAN images, thereby strictly limiting the upper bound of cross-modal retrieval precision [6]. On the other hand, the significant domain shift between different modalities critically impedes feature space alignment at both spatial and semantic levels [21], [26]. This shift manifests as a complex "spatial–spectral" distribution discrepancy: PAN images rely on grayscale structural textures, whereas MS images prioritize multiband spectral reflectance [5]. This divergence results in a disjointed feature space where spatial structures and semantic contexts fail to correspond across modalities. Consequently, without robust alignment strategies to bridge this gap, the extracted features struggle to achieve spatial-

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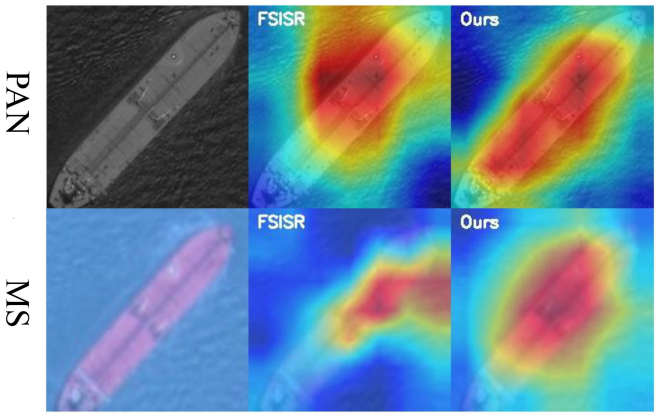


Fig. 1. Visual comparison of class activation mapping (Grad-CAM) in complex maritime scenarios. The top row (PAN) and bottom row (MS) illustrate that existing methods (FSISR) suffer from fragmented attention and background distraction, while our FG-MD achieves precise target coverage.

semantic consistency. Although methods such as SIDHCNN [5] and FSISR [6] incorporate similarity constraints, they lack effective mechanisms to calibrate these heterogeneous distributions under complex interference, ultimately leading to retrieval failure. As intuitively illustrated in Fig. 1, existing methods exhibit fragmented activation responses due to the spatial misalignment induced by resolution loss and sea clutter. Furthermore, the efficiency-accuracy trade-off remains a critical bottleneck for practical deployment. Strictly controlling model complexity under limited computational resources while maintaining fine-grained feature representation is an unresolved challenge, particularly given the growing demand for on-board edge computing. Therefore, designing a lightweight model that simultaneously bridges the modality gap and preserves fine-grained semantics without compromising retrieval efficiency is of substantial practical significance.

To address these challenges, we propose the frequency-guided mutual distillation (FG-MD) network, a lightweight architecture that innovatively decouples cross-modal retrieval into two stages: global feature perception and dual-dimensional feature space alignment. Specifically, we design a fine-grained feature enhancement and denoising framework comprising the frequency-domain full-spectrum perceiver (FDFSP) module and an adaptive dynamic filter (ADF). To address the loss of detail in high-level semantic features, the FDFSP innovatively incorporates a frequency-domain modulation mechanism. Unlike spatial convolution, which focuses on local intensities, FDFSP leverages the spectral properties of the FFT to facilitate full-spectrum awareness [27], effectively emphasizing the perception of high-frequency components (e.g., edges and textures) that are typically smoothed out by convolutional operations. Subsequently, the ADF utilizes dynamic convolution to construct content-adaptive filters. By dynamically recalibrating feature weights based on global context, the ADF functions as a semantic gate to precisely suppress background clutter while preserving target-relevant responses. Crucially, this refinement process establishes a clean and robust feature foundation. Building on this, we introduce a dual-attention mutual distillation (DAMD) strategy [28] to achieve cross-modal alignment within a dual-dimensional feature space. Specifically, unlike simple denoising or similarity constraints, DAMD employs a symmetric collaborative learning mechanism to progressively rectify discrepancies across spatial and semantic dimensions, thereby ensuring high-level consistency in the spatial-semantic dual-dimensional feature space with zero inference overhead. Furthermore, FG-MD employs MobileNetV2 [29] as a lightweight backbone to facilitate end-to-end training. Experiments on the MRSSID benchmark [7] demonstrate that our model achieves state-of-the-art results across all settings with approximately 22-M parameters, striking an optimal balance between retrieval accuracy and inference efficiency.

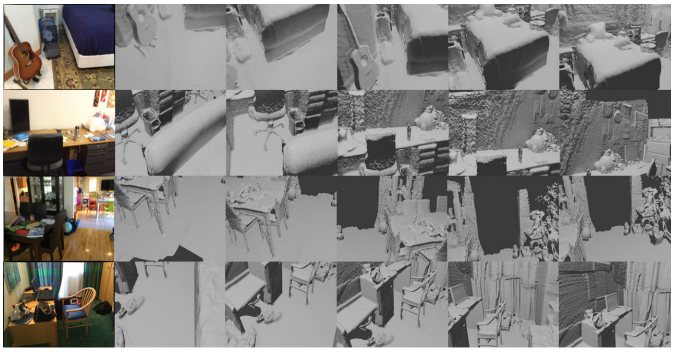


Fig. 2. Overview of the constructed IGS benchmark. The dataset features significant modality gaps between high-fidelity RGB images and degraded point cloud projections, along with challenging conditions such as severe occlusions and unstructured clutter typical of GPS-denied UAV environments.

To further validate the model's generalization capability, we constructed the indoor geometric structure (IGS) benchmark, specifically designed for UAV indoor reconnaissance in GPS-denied environments. This dataset projects core remote sensing challenges onto the UAV 3-D observation domain, characterized by three key aspects as follows.

1) *Modality Heterogeneity:* As visualized in Fig. 2, IGS utilizes high-fidelity RGB images and degraded point cloud projections to analogize and reproduce the modality gap between PAN and MS images.

2) *Observation Constraints:* Unlike sparse top-down views, indoor perspectives are filled with dense occlusions and unstructured clutter, significantly increasing the difficulty of feature decoupling.

3) *Evaluation Standards:* Addressing UAV high mobility, the benchmark requires maintaining geometric discriminability under drastic viewpoint rotations. This configuration realistically simulates critical scenarios like indoor rescue, effectively filling the gap for challenging benchmarks in the field of CMRSIR. Experiments indicate that our method outperforms FSISR by over 10% on IGS, demonstrating cross-domain robustness ranging from satellite remote sensing to close-range UAV reconnaissance.

The main contributions of this work are summarized as follows.

1) We propose FG-MD, a lightweight framework that achieves a superior tradeoff between efficiency and accuracy. With only 22-M parameters, it establishes a

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new state-of-the-art performance (0.8654 mAP) on the MRSSID benchmark, significantly outperforming heavy baselines with minimal computational cost.

2) We introduce a frequency-guided denoising mechanism comprising the FDFSP and adaptive dynamic filter (ADF). These modules synergistically enhance global structural perception and suppress high-frequency clutter via dynamic filtering, effectively compensating for the limited capacity of lightweight backbones.
3) We devise a DAMD strategy for precise cross-modal alignment. By enforcing symmetric consensus on semantic channels and spatial focus, this mechanism eliminates background interference and boosts feature robustness without incurring any extra inference overhead.
4) We construct and release IGS, a challenging benchmark for asymmetric visual localization. Featuring severe geometric distortions in GPS-denied environments, IGS serves as a rigorous testbed to evaluate the cross-domain generalization capability of lightweight models beyond standard remote sensing scenarios.

II. Related Work

A. From Real-Valued Features to Cross-Modal Metrics

Early approaches to remote sensing image retrieval predominantly relied on hand-crafted low-level features, such as SIFT and HOG [20], [30], [31], [32], [33]. However, these shallow descriptors often fail to effectively characterize the high-level semantic information inherent in complex remote sensing scenes. With the advent of deep learning, convolutional neural networks (CNNs) have emerged as the mainstream paradigm, owing to their powerful feature extraction capabilities [29], [34], [35]. For instance, Xiong et al. [7] introduced an attention mechanism combined with a multitask learning strategy to extract salient discriminative features from remote sensing images effectively. Similarly, Li et al. [13] enhanced feature representation in complex scenes by integrating CNNs with the vector of locally aggregated descriptors (VLADs).

However, restricted by single data sources, existing methods fail to leverage heterogeneous sensors. Consequently, CMIR has emerged to exploit complementary information, including the spatial details of PAN images and the spectral information of MS images [21], [26]. The core challenge lies in bridging the modality gap caused by distinct imaging mechanisms [24]. While recent approaches like fusion transformers and IFSN [7] achieve precise matching using dense real-valued features, their high dimensionality incurs prohibitive storage costs and latency, hindering real-time performance [6].

B. Evolution of Deep Hashing Paradigms

To circumvent the efficiency bottlenecks associated with real-valued features, hashing retrieval technology maps high-dimensional data into compact binary codes [18], [19], [36], [37], [38], thereby enabling rapid Hamming distance computation and low storage consumption [39], [40], [41], [42], [43]. The evolution of hashing techniques has transitioned from data-independent methods, such as locality sensitive hashing (LSH) [44], to data-dependent learning-to-hash approaches [45], [46], [47], [48], [49]. In the single-modal domain, DPSH [50] pioneered the utilization of pairwise labels to optimize feature representation and hash functions simultaneously. Meanwhile, HashNet [51] introduced a continuation strategy with nonsmooth activations to address discrete optimization difficulties. Recently, transformer-based methods, such as HashFormer [52], have also been employed to capture long-range dependencies in remote sensing images.

C. Cross-Modal Remote Sensing Hashing

With the increasing availability of multisource earth observation data, multimodal learning has been successfully applied to diverse tasks, such as classification [53], semantic segmentation [54], and foundation model pretraining [55]. Within this context, CMRSIR has emerged as a cornerstone of modern intelligent systems. To facilitate efficient large-scale retrieval, cross-modal hashing has emerged as a predominant paradigm by projecting heterogeneous data into a shared Hamming space while preserving semantic affinities. Early exploratory works, such as CCA [30], CMFH [20], ITQ [19], SH [18], SCM [31], and LSSH [32], primarily utilized linear projections or matrix factorization to find common latent subspaces. However, these shallow methods rely on hand-crafted descriptors and struggle to capture the complex, nonlinear correlations inherent in multisource remote sensing imagery.

End-to-end deep hashing frameworks, such as DCMH [21], CMHH [26], SSAH [22], and others [24], [25], [56], integrate feature extraction with coding to enhance retrieval accuracy. To address the nondifferentiable nature of hashing, these methods typically employ continuous relaxation combined with quantization and ranking losses [23] to optimize discriminative binary codes. In the remote sensing domain, architectures like SIDHCNN [5] and FSISR [6] incorporate semantic alignment to bridge the PAN-MS modality gap. However, their reliance on parameter-heavy backbones (e.g., ResNet-50) incurs high computational costs, severely hindering real-time deployment on edge platforms.

D. Comparative Analysis of FG-MD and Existing Paradigms

Traditional cross-modal paradigms predominantly rely on unidirectional teacher–student distillation or static frequency filtering, which often struggle to reconcile fine-grained representation with semantic consistency. Although recent advancements, such as self-supervised distillation [57] and multiteacher strategies [58], have improved feature transfer, they are frequently limited by the inherent modality gap and spectral bias in remote sensing. In contrast, the FG-MD framework introduces a symmetric collaborative learning strategy. By enabling PAN and MS streams to supervise each other within frequency-aligned latent spaces, FG-MD effectively bridges the domain shift without requiring heavy-duty teacher models.

Furthermore, existing frequency-domain methods, such as FDNet [59] and multiscale spatial-frequency networks [60], have explored shifting feature extraction from the spatial to the spectral domain. However, these approaches often

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employ static transforms that inadvertently amplify environmental noise, such as sea clutter. To address this issue, the FDFSP module performs content-adaptive frequency modulation. By dynamically coupling high-frequency structural cues with semantic channels, FDFSP selectively enhances fine-grained details relevant to retrieval while strictly suppressing modality-specific clutter, thereby ensuring robust performance in complex heterogeneous scenarios.

Finally, while modern cross-modal retrieval frameworks, such as FSISR [6], have achieved substantial improvements through feature semi-interaction, they typically depend on computationally intensive backbones, which restricts their deployment on resource-constrained edge platforms. Unlike prior hashing methods that focus predominantly on spatial-domain learning, FG-MD integrates adaptive frequency modulation with a DAMD strategy. This synergy allows the proposed model to achieve superior retrieval accuracy while maintaining highly competitive model efficiency. By explicitly enhancing structural information with zero inference overhead, FG-MD offers an optimal tradeoff between performance and cross-domain generalization.

III. Method

CMRSIR aims to identify geographically consistent samples in one modality using queries from another. For clarity, we formulate the problem using PAN images $\mathcal{X}^p = \{x_i^p\}_{i=1}^N$ as queries to retrieve MS images $\mathcal{X}^m = \{x_i^m\}_{i=1}^N$, where N represents the number of samples. We construct feature extraction backbones $\sigma_{ms}(\cdot)$ and $\sigma_{pan}(\cdot)$, and learn modality-specific mapping functions, denoted as $\mathcal{F}_{ms}$ and $\mathcal{F}_{pan}$. Subsequently, leveraging a deep hashing framework, these heterogeneous inputs are projected into a shared K-bit binary Hamming space $\{-1, 1\}^K$. Within this latent space, the retrieval process is governed by the preservation of semantic similarity, ensuring that the Hamming distances between binary codes of positive pairs representing the same geographical target are minimized, whereas those of negative pairs are maximized. Ultimately, the top-K samples exhibiting the smallest Hamming distances are selected as the final retrieval results.

A. Problem Analysis and Motivation

In CMRSIR tasks, deep hashing-based methods encounter a fundamental impediment: the severe asymmetry between spatial structure and semantic fidelity. Specifically, on one hand, the pursuit of high-level semantic features inevitably entails a significant loss of spatial information. Taking PAN-MS retrieval as an example, let $\sigma(\cdot)$ denote the mapping of MS (x^m) and PAN (x^p) inputs toward a latent semantic prototype z_{sem}. As the network extracts high-level semantics, the feature representations inevitably diverge due to modality-specific degradation mechanisms

$$\begin{aligned} \sigma_m(x^m) &\approx z_{sem} - \Delta_{spatial} \\ \sigma_p(x^p) &\approx z_{sem} + \epsilon_{clutter}. \end{aligned} \tag{1}$$

Here, $\Delta_{spatial}$ signifies the inherent deficiency of fine-grained spatial details in high-level feature extraction, while $\epsilon_{clutter}$ represents the additive interference from complex background clutter. Standard optimization objectives effectively degenerate into minimizing the residual error $\|\epsilon_{clutter} + \Delta_{spatial}\|$. Without explicit spatial structure maintenance, the model is compelled to compromise, attempting to align spatially deficient representations with those corrupted by noise, which leads to severe feature misalignment (as shown in Fig. 1).

On the other hand, a paradox exists between achieving model efficiency and ensuring cross-modal feature space alignment. Rectifying the deviation in (1) typically necessitates computationally expensive mechanisms, such as heavy attention modules for global context or complex reconstruction modules, which are prohibitive for spaceborne platforms. However, standard lightweight architectures lack the capacity to capture long-range dependencies for spatial compensation or to perform the dynamic calibration required to filter local noise. This limitation highlights the necessity of integrating frequency-domain modulation and content-adaptive filtering frameworks without sacrificing end-to-end efficiency.

B. Overview

Fig. 3 illustrates the overall architecture of the proposed FG-MD framework. Designed as a symmetric dual-stream network, our method primarily comprises three key stages: lightweight feature extraction, frequency-guided refinement, and hash generation with mutual optimization.

Specifically, given an MS input x^m and a PAN input x^p, we employ two parallel MobileNetV2 backbones to extract preliminary spatial representations. Let $\sigma^s(\cdot)$ denote the backbone feature extractor. The initial deep features F_{feat}^s for modality $s \in \{m, p\}$ are computed as

$$F_{\text{feat}}^s = \sigma^s(x^s) \in \mathbb{R}^{h \times w \times c}. \tag{2}$$

Subsequently, the features enter a denoising framework to mitigate degradation. Specifically, the FDFSP leverages FFT-based global interactions to compensate for spatial deficiencies, followed by the ADF, which employs dynamic weight recalibration to precisely suppress background clutter while preserving semantics. This refinement process is formulated as

$$F_{\text{refined}}^s = \mathcal{H}_{\text{ADF}}\left(\mathcal{H}_{\text{FDFSP}}\left(F_{\text{feat}}^s\right)\right) + F_{\text{feat}}^s. \tag{3}$$

In the projection stage, the refined representations are aggregated by a DataFusion module and projected into the target Hamming space via a Hash Head to generate compact binary codes. During training, we employ a symmetric mutual distillation strategy that leverages spatial-semantic dual-dimensional distillation to enforce cross-modal feature alignment. This approach ensures cross-modal consistency without relying on auxiliary attention mechanisms. The complete step-by-step training procedure, detailing the interaction between the distillation strategy and refinement modules, is summarized in Algorithm S1 of the Supplementary Material.

C. Frequency-Domain Full-Spectrum Perceiver

To compensate for the spatial deficiencies ($\Delta_{spatial}$) inherent in high-level semantics and effectively emphasize the

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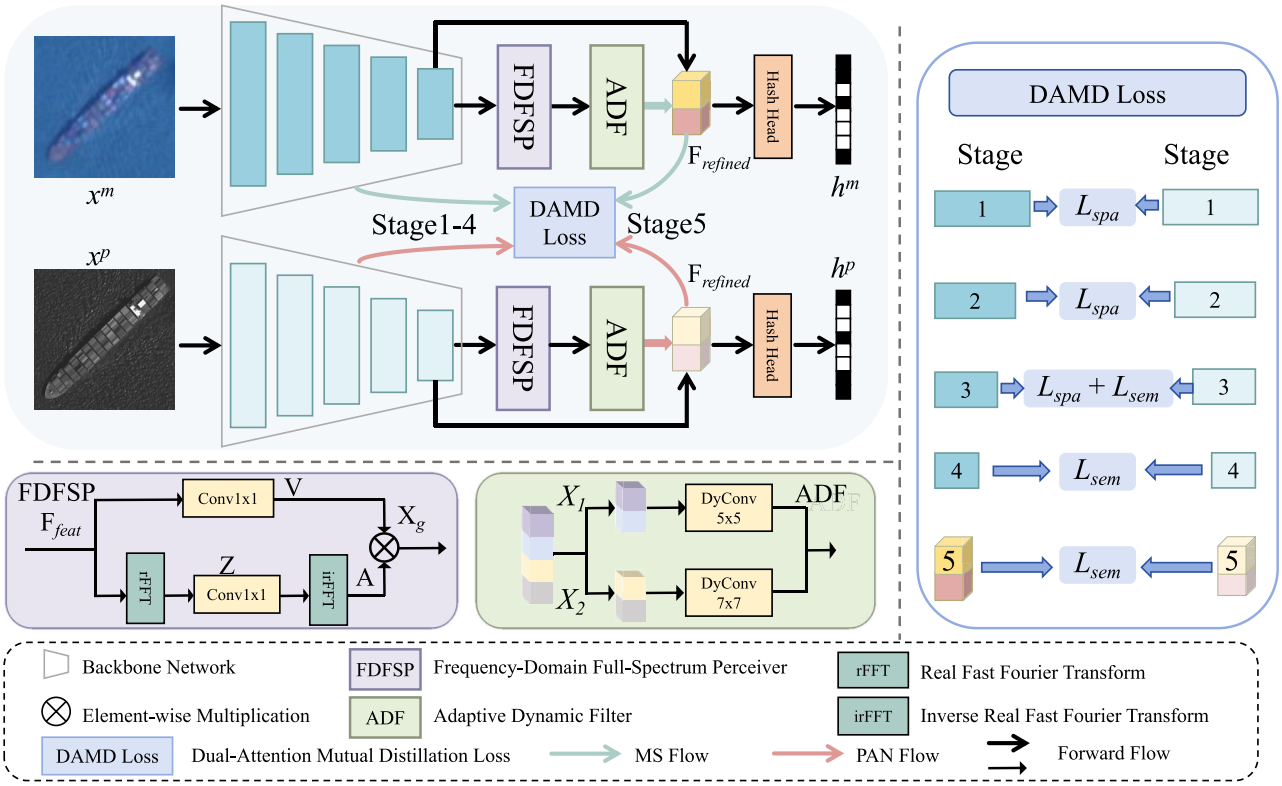


Fig. 3. Overall architecture of the proposed FG-MD framework. The framework follows a symmetric dual-stream design to efficiently process MS (x^m) and PAN (x^p) inputs under edge-device constraints. A lightweight MobileNetV2 backbone is adopted for feature extraction, followed by the FDFSP and ADF to efficiently enhance global perception and suppress sea clutter. Finally, the DAMD strategy aligns and refines cross-modal representations via collaborative optimization.

fine-grained texture details typically attenuated during downsampling, we introduce the FDFSP. As shown in Fig. 2, the FDFSP employs a dual-path structure to implement a frequency-domain modulation mechanism. Let $\mathbf{X}$ denote the input features (representing F^s_{feat}). In the spatial path, a 1×1 convolution generates a local feature projection $\mathbf{V}$

$$\mathbf{V} = \text{Conv}_{1\times1}(\mathbf{X}) \in \mathbb{R}^{h\times w\times c}. \tag{4}$$

Parallel to this, the spectral path leverages the global interaction capabilities of the FFT to capture long-range dependencies. The features are projected into the frequency domain using the 2-D real fast Fourier transform (rFFT)

$$\mathbf{Z} = \texttt{rFFT}(\mathbf{X}) \in \mathbb{C}^{h\times(\frac{w}{2}+1)\times c}. \tag{5}$$

In the spectral domain, a complex-valued convolution modulates the amplitude and phase components, effectively performing a global circular convolution that transcends local receptive fields. The modulated spectrum is then transformed back to the spatial domain via the inverse real FFT (irFFT) to produce the global weight map $\mathbf{A}$

$$\mathbf{A} = \texttt{irFFT}\left(\text{Conv}_{1\times1}(\mathbf{Z})\right) \in \mathbb{R}^{h\times w\times c}. \tag{6}$$

The global structural descriptors extracted by the spectral path then gate the local feature responses. The spectral modulator $\mathbf{A}$ acts as a global filter, recalibrating the spatial projection $\mathbf{V}$ through element-wise multiplication ($\otimes$). This process injects global context into local features, ensuring the restoration of target integrity. The FDFSP output, $\mathbf{X}_g$, is defined as

$$\mathbf{X}_g = \mathbf{V} \otimes \mathbf{A}. \tag{7}$$

This modulation mechanism is theoretically grounded in the spectral properties of natural images, serving as a direct countermeasure to the spatial deficiency $\Delta_{spatial}$ formulated in (1) [61]. According to phase congruency theory, geometric structures—such as edges and ship boundaries—are primarily encoded in the phase information [62]. Although pixel-level intensities vary between MS and PAN sensors, the underlying geometric semantics remain consistent. By performing convolution in the complex domain, the learnable weights of the FDFSP implicitly adjust the phase distribution to enhance these modality-invariant structural cues [63], [64], effectively reversing the structural erosion characterized by $\Delta_{spatial}$. Simultaneously, the magnitude of the complex weights acts upon the amplitude spectrum to recalibrate the spectral energy distribution [65]. Since semantic targets typically dominate low-frequency components while noise appears as high-frequency scatter, this module adaptively emphasizes low-frequency responses associated with target integrity. This unified frequency-domain processing facilitates simultaneous structural enhancement and clutter suppression, establishing a robust foundation for subsequent feature alignment.

D. Adaptive Dynamic Filter (ADF)

While FDFSP focuses on global structure, the feature maps may still contain local noise caused by complex sea states,

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such as turbulent waves and specular reflections. Standard static convolutions use fixed parameters that struggle to adapt to dynamic maritime clutter. To address this, we introduce the ADF, which uses content-conditioned dynamic convolutions for pixel-level filtering.

As shown in Fig. 2, ADF partitions the input features $\mathbf{X}_g$ (from FDFSP) along the channel dimension into two groups, $\mathbf{X}_1$ and $\mathbf{X}_2$. This split design handles multiscale noise distributions. For the $\mathbf{X}_1$ branch, we generate instance-specific kernel weights $\mathbf{W}_{dyn}$ rather than using fixed filters. We use global average pooling (GAP) followed by a lightweight multilayer perceptron (MLP) to capture channel-wise context:

$$\mathbf{W}_{dyn} = \mathrm{MLP}\left(\mathrm{GAP}\left(\mathbf{X}_1\right)\right). \tag{8}$$

We then use $\mathbf{W}_{dyn}$ for depthwise convolution on $\mathbf{X}_1$. By adjusting filter responses based on local content, ADF distinguishes target pixels from background clutter. To ensure multiscale perception, the second branch ($\mathbf{X}_2$) uses a different kernel size (e.g., 7×7). Finally, we apply a channel shuffle to enable information exchange between groups, followed by a 1×1 convolution to generate the final features.

This adaptive spatial filtering complements the global spectral modulation of FDFSP, providing a dual-domain (spectral–spatial) solution to environmental interference in complex maritime environments.

E. Dual-Attention Mutual Distillation

Bridging the extensive modality gap represents a fundamental challenge in cross-modal retrieval, often leading to suboptimal manifold alignment due to the intrinsic heterogeneity of sensor data. Unlike traditional approaches that rely on computationally expensive auxiliary modules to forcibly reconcile these distributions, our goal is to achieve precise feature space alignment without incurring any additional computational burden during inference. To this end, we devise a DAMD strategy.

DAMD establishes a synergistic loop with the aforementioned denoising framework. While FDFSP and ADF provide purified feature foundations, DAMD acts as a zero-overhead collaborative regularizer to further decouple modality-specific noise. By employing a symmetric collaborative learning mechanism, this strategy facilitates the reciprocal assimilation of complementary strengths during training. Specifically, it integrates the rich semantic cues inherent in MS imagery with the fine-grained geometric details from PAN imagery, thereby ensuring strict consistency across the spatial-semantic dual-dimensional feature space.

To achieve hierarchical alignment, we implement channel-semantic alignment in deep stages ($k \in \{3, 4, 5\}$) to forge high-level semantic consensus, and spatial-focus alignment in shallow layers ($k \in \{1, 2, 3\}$) to synchronize fine-grained geometric attention. Notably, the third stage serves as a pivotal transitional bridge, simultaneously imposing dual constraints to span the gap between primitive textures and abstract concepts.

1) Channel-Semantic Alignment: Deep feature channels primarily capture high-level discriminative patterns. To synchronize these semantic distributions, we implement an attention-guided feature distillation (AFD) strategy. Given the feature map $\mathbf{F}_{cp}^k$ from the counterpart stream at stage k, we derive a channel importance vector $\boldsymbol{\rho}_{ch}$ to identify informative semantics. This vector is computed via GAP, followed by a bottleneck MLP and a Sigmoid function, normalized across the channel dimension:

$$\boldsymbol{\rho}_{ch}\left(\mathbf{F}_{cp}^k\right) = \frac{\sigma\left(\mathrm{MLP}\left(\mathrm{GAP}\left(\mathbf{F}_{cp}^k\right)\right)\right)}{\sum_c \sigma\left(\mathrm{MLP}\left(\mathrm{GAP}\left(\mathbf{F}_{cp}^k\right)\right)\right)_c + \epsilon} \tag{9}$$

where ϵ is a constant for numerical stability. The semantic distillation loss, $\mathcal{L}_{sem}$, is formulated as the attention-weighted mean squared error (mse) between the normalized features of the current stream ($\tilde{\mathbf{F}}_{curr}$) and the counterpart stream ($\tilde{\mathbf{F}}_{cp}$)

$$\mathcal{L}_{sem}\left(\mathbf{F}_{curr}, \mathbf{F}_{cp}\right) = \sum_{k \in \mathcal{K}_{sem}} \alpha_k \cdot \left\| \boldsymbol{\rho}_{ch}\left(\mathbf{F}_{cp}^k\right) \odot \left(\tilde{\mathbf{F}}_{curr}^k - \tilde{\mathbf{F}}_{cp}^k\right) \right\|_2^2 \tag{10}$$

where $\mathcal{K}_{sem} = \{3, 4, 5\}$ denotes the set of indices for deep stages focused on semantic alignment. Here, $\tilde{\mathbf{F}}$ denotes the L_2-normalized feature map. This constraint compels the current network to prioritize the reconstruction of semantically critical channels highlighted by its counterpart.

By focusing on these late stages, $\mathcal{L}_{sem}$ functions as a semantic filter. It constrains the network to prioritize the synchronization of high-level discriminative patterns (where ρ_{ch} is large) while permitting flexibility in channels that capture modality-specific textures. Consequently, this constraint forces the current stream to achieve a high-level semantic consensus with its counterpart, ensuring that the learned hash codes are robust to spectral variations.

2) Spatial-Focus Alignment: To mitigate maritime clutter, we align the spatial attention of both streams. Analogous to the semantic module, a spatial attention map $\boldsymbol{\rho}_{sp}$ is derived from the counterpart features using a 3×3 convolution layer

$$\boldsymbol{\rho}_{sp}\left(\mathbf{F}_{cp}^k\right) = \frac{\sigma\left(\mathrm{Conv}\left(\mathbf{F}_{cp}^k\right)\right)}{\sum_{h,w} \sigma\left(\mathrm{Conv}\left(\mathbf{F}_{cp}^k\right)\right)_{h,w} + \epsilon}. \tag{11}$$

This map highlights salient spatial regions (e.g., ships) while suppressing background noise. The spatial loss $\mathcal{L}_{spa}$ leverages this map to weight the feature alignment

$$\mathcal{L}_{spa}\left(\mathbf{F}_{curr}, \mathbf{F}_{cp}\right) = \sum_{k \in \mathcal{K}_{spa}} \beta_k \cdot \left\| \boldsymbol{\rho}_{sp}\left(\mathbf{F}_{cp}^k\right) \odot \left(\tilde{\mathbf{F}}_{curr}^k - \tilde{\mathbf{F}}_{cp}^k\right) \right\|_2^2 \tag{12}$$

where $\mathcal{K}_{spa} = \{1, 2, 3\}$ corresponds to the shallow stages, and $\boldsymbol{\rho}_{sp}(\mathbf{F}_{cp}^k) \in \mathbb{R}^{1 \times H \times W}$ is a 2-D spatial weight map applied via channel-wise broadcasting.

By concentrating on these initial stages, $\mathcal{L}_{spa}$ acts as a spatial gatekeeper. It enforces strict feature alignment in regions with high saliency (where ρ_{sp} is large) while allowing for divergence in background clutter. Consequently, the network prioritizes modality-invariant targets and effectively disregards sensor-specific background interference.

3) Mutual Optimization: We adopt a symmetric collaborative learning paradigm where both branches function as mutual guides. The total distillation objective $\mathcal{L}_{dist}$ incorporates bidirectional losses, where each modality guides the other using its generated attention maps

$$\mathcal{L}_{dist} = \sum_{s \in \{m,p\}} \left(\mathcal{L}_{sem}\left(\mathbf{F}_s, sg\left[\mathbf{F}_{\bar{s}}\right]\right) + \mathcal{L}_{spa}\left(\mathbf{F}_s, sg\left[\mathbf{F}_{\bar{s}}\right]\right)\right) \tag{13}$$

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where $\bar{s}$ denotes the counterpart modality, and $sg[\cdot]$ represents the stop-gradient operator. This symmetric constraint ensures that the rich spectral cues from the MS branch and the precise geometric details from the PAN branch implicitly guide the feature refinement of the counterpart backbone.

F. Optimization Objectives

The proposed FG-MD framework is optimized end-to-end to jointly learn modality-specific feature representations and discrete hash codes. Following the standard protocol established in FSISR [6], we employ a composite objective to ensure that the learned Hamming space is both semantically discriminative and structurally robust. Specifically, we incorporate a cross-entropy classification loss ($\mathcal{L}_{cls}$) to preserve categorical information, a quantization loss ($\mathcal{L}_{quant}$) to minimize the discretization error between continuous embeddings and binary codes, and a triplet ranking loss ($\mathcal{L}_{tri}$) to maintain relative similarity orders. Upon integrating the proposed mutual distillation regularization, the total objective function $\mathcal{L}_{total}$ is formulated as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \mathcal{L}_{quant} + \mathcal{L}_{tri} + \lambda_{dist}\mathcal{L}_{dist} \tag{14}$$

where $\mathcal{L}_{dist}$ represents the dual-attention mutual learning objective defined in Section III-E, and λ_{dist} is the hyperparameter balancing its contribution to the collaborative optimization process. Detailed mathematical formulations and the structural significance of each loss component are provided in Section I of the Supplementary Material.

IV. Experiments

A. Experimental Settings

1) Datasets: In addition to standard benchmarking on the widely used MRSSID dataset and VAIS dataset, we employ the newly constructed IGS dataset to further verify the robustness of FG-MD in cross-modal view retrieval under complex heterogeneous scenarios.

MRSSID Dataset: This is a standard benchmark dataset for cross-modal remote sensing ship image retrieval. It comprises 2865 pairs of MS and PAN images, covering six distinct ship categories: destroyers, littoral combat ships, combat boats, bulk carriers, container ships, and oil tankers. Following the standard protocol [7], we randomly partition the dataset into training and testing sets.

VAIS Dataset: To assess generalization, the performance of FG-MD is further evaluated on the VAIS benchmark [66]. This dataset consists of paired visible (VIS) and infrared (IR) ship images categorized into six classes. To ensure a rigorous evaluation, the official training and testing split provided by the VAIS dataset is adopted, and identical hyperparameter configurations from the MRSSID experiments are maintained. This extension serves to verify the robustness of the model against substantial modality gaps between thermal and optical spectra under consistent experimental constraints.

CBRSIR_VS Dataset: To further evaluate the robustness of FG-MD under extreme modality discrepancies, experiments are conducted on the CBRSIR_VS dataset [17]. This cross-modal benchmark is developed based on the EuroSAT dataset and comprises 26 901 pairs of optical (VHR) and synthetic aperture radar (SAR) images across ten land-use categories. The VHR images, acquired from Google Earth and ArcGIS, feature a spatial resolution of 1 m and a dimension of 256×256 pixels. In contrast, the SAR images are collected from Sentinel-1 (VV polarization) with a spatial resolution of 10 m and a dimension of 64×64 pixels. Following the evaluation protocol in [60], 100 images per category are randomly selected for the test set, while the remaining images constitute the retrieval set.

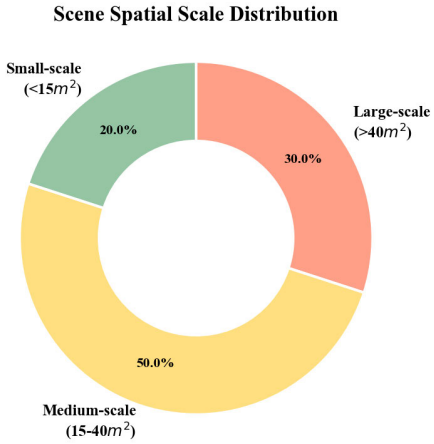


Fig. 4. Distribution of scene spatial scales in the IGS benchmark. Based on typical architectural layouts, scenes are categorized into small-scale (< 15 m^2), medium-scale (15–40 m^2), and large-scale (> 40 m^2) groups.

IGS Benchmark: To simulate the rigorous observation constraints of close-range UAV reconnaissance in GPS-denied environments, we constructed the IGS benchmark utilizing ten high-fidelity 3-D point cloud models. This dataset serves as a challenging isomorphic extension of standard remote sensing tasks, projecting core Earth observation challenges into the 3-D near-field domain. Transcending traditional 2-D augmentations, we introduce a "multiview trajectory sampling" strategy to realistically model 6-DoF flight kinematics and severe geometric distortions. This process generates a total of 2100 heterogeneous image pairs across 100 unstructured scenes, explicitly reproducing the modality gap and viewpoint variability inherent to complex reconnaissance missions. The details are as follows.

a) *Data generation and viewpoint sampling:* For each 3-D scene, a high-resolution RGB reference map (I_{ref}) is rendered at a 2.5-m distance. To evaluate geometric robustness, we sample ten discrete query views with angular offsets of up to $\pm 30^\circ$ by simulating a UAV's rotational trajectory. The complete generation process and sampling pseudocode are detailed in Algorithm S2 of the Supplementary Material.

b) *Scene distribution:* The dataset comprises 100 scenes exhibiting a wide range of spatial scales. As shown in Fig. 4, based on typical indoor architectural layouts, we empirically categorize the scenes into three scales: small-scale (< 15 m^2, 20%), medium-scale (15–40 m^2, 50%), and large-scale (> 40 m^2, 30%). This distribution ensures robustness across diverse geometric complexities.

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c) *Train/test split:* To prevent data leakage, we adopt a scene-level splitting strategy. The 100 scenes are randomly divided into a training set (70 scenes) and a testing set (30 scenes) with no overlapping environments.

2) Evaluation Metrics: To comprehensively assess the retrieval performance of the proposed FG-MD framework, we employ the mean average precision at rank R (mAP@R) as the primary evaluation metric.

a) Retrieval performance: The average precision (AP) for a specific query image q at a cutoff rank R is defined as

$$AP@R(q) = \frac{1}{N_R} \sum_{k=1}^{R} P(k) \times \delta(k) \tag{15}$$

where N_R denotes the total number of relevant images within the top-R retrieved results (or the total number of ground-truth relevant images, depending on the protocol), $P(k)$ represents the precision at rank k, and $\delta(k)$ is an indicator function that equals 1 if the kth retrieved image is relevant to the query, and 0 otherwise.

Based on this, the mAP@R is calculated by averaging the AP scores over the entire query set $\mathcal{Q}$

$$mAP@R = \frac{1}{|\mathcal{Q}|} \sum_{i=1}^{|\mathcal{Q}|} AP@R(q_i) \tag{16}$$

where $|\mathcal{Q}|$ denotes the total number of query images.

In our experiments, the value of R is adapted to the scale and characteristics of the datasets. For standard public benchmarks, we set $R = 86$ following established protocols. For our self-constructed IGS dataset, considering the specific application requirements, we report performance at multiple cutoffs, specifically $R \in \{1, 3, 5\}$ (denoted as mAP@1, mAP@3, and mAP@5), to evaluate precision at strictly limited retrieval depths.

b) Model efficiency and scalability: To assess the practical utility of FG-MD in resource-constrained environments, we benchmark efficiency across two dimensions. First, model complexity and inference speed are evaluated via the number of parameters (Params), storage size (Size), peak video memory usage (VRAM), and floating-point operations (FLOPs). Inference speed (FPS) is measured across CPU, GPU, and embedded Jetson NX platforms to simulate edge deployment latency. Second, to verify robustness against high-resolution inputs, we analyze scalability across resolutions from 128^2 to 1024^2. FLOPs and VRAM growth rates are monitored to quantify the computational stability of FG-MD under varying observation scales.

3) Implementation Details: The proposed framework is implemented using the PyTorch library on a single NVIDIA RTX 5880 Ada GPU, with the MobileNetV2 backbone initialized via ImageNet pretrained weights to facilitate faster convergence. For data preprocessing, input images are resized to 256×256 pixels, from which 224×224 patches are randomly cropped during training for data augmentation and center-cropped during testing. Model optimization is conducted over 200 epochs with a batch size of 128 using the Adam optimizer, where the momentum parameters β_1 and β_2 are configured to 0.9 and 0.999, respectively. The initial learning rate is set to 1×10^{-4} and decreased by a factor of 0.9 every 10 epochs using a step decay policy. Based on empirical validation, the loss weighting coefficients are specified as $\lambda_{tri} = 1$ and $\lambda_{dist} = 0.7$, while the retrieval performance is evaluated across various hash code lengths $K \in \{16, 24, 32, 48, 64\}$ bits to ensure a comprehensive comparison.

B. Comparison With State-of-the-Art Methods

1) Comparison With Baseline Methods: To evaluate the proposed FG-MD, we conducted comparative experiments on the MRSSID dataset with hash codes ranging from 16 to 64 bits. The quantitative results (mAP) are summarized in Table I.

As summarized in Table I, deep hashing methods exhibit a distinct advantage over shallow models due to their end-to-end optimization capabilities. Notably, FG-MD consistently establishes a new state-of-the-art performance across all bit lengths. In the 24-bit PAN → MS task, FG-MD achieves a peak mAP of 0.8654, which significantly outperforms the specialized SOTA method FSISR (0.8204). Furthermore, to provide a more comprehensive evaluation as suggested, the proposed model is compared against several recent advanced architectures. FG-MD demonstrates superior performance over CCR-Net [53] (0.7524), and the representative Mamba-based baseline MambaHash [67] (0.7268). Remarkably, the proposed model even exceeds the performance of the massive ViT-Large [21] (0.8327). This validates that the frequency-domain denoising (FDFSP) and ADF of FG-MD are more effective than brute-force global attention or general-purpose backbones for resolving fine-grained RS spectral discrepancies.

Table II details the fine-grained class-wise performance. FG-MD demonstrates consistent superiority over FSISR, particularly on categories with high interclass similarity (e.g., oil tankers and bulk carriers) that are easily confused. Notably, for Oil Tankers in the MS → PAN task (32-bit), FG-MD achieves a 10.1% absolute gain (0.7195–0.8205). This robustness indicates that the FDFSP module successfully captures subtle structural cues often missed by spatial-only convolutions, while the ADF precisely isolates the target subject from background interference.

Beyond the MS-PAN task, we evaluate the robustness of FG-MD on the VAIS dataset, which focuses on IR-to-VIS (IR-VIS) retrieval. As detailed in Table III, FG-MD maintains a significant margin over current SOTA methods. Specifically, in the VIS → IR task, FG-MD achieves an mAP of 0.7801 at 24 bits, outperforming the strongest competitor FSISR (0.6891) by a substantial margin of 9.10%. This performance gap on IR-VIS imagery, where modality discrepancy is typically more pronounced than in MS-PAN, underscores the efficacy of our DAMD in capturing modality-invariant features across diverse spectral ranges.

Finally, to evaluate FG-MD under extreme modality discrepancies, we conduct experiments on the CBRSIR_VS dataset (Table IV), which exhibits a substantial modality gap between VHR and SAR imagery. As illustrated in Table IV, FG-MD consistently achieves the superior mAP across all

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TABLE I

COMPARISON OF MAP OF HAMMING RANKING FOR DIFFERENT BITS ON CROSS-MODAL RS SHIP IMAGE RETRIEVAL

Architecture	Method	MS → PAN					PAN → MS				
		16 bits	24 bits	32 bits	48 bits	64 bits	16 bits	24 bits	32 bits	48 bits	64 bits
Shallow	CMFH [20]	0.1264	0.1207	0.1250	0.1319	0.1273	0.1391	0.1326	0.1475	0.1540	0.1510
	IMH [33]	0.1358	0.1332	0.1386	0.1422	0.1388	0.1295	0.1250	0.1274	0.1322	0.1323
	LSSH [32]	0.1310	0.1331	0.1362	0.1406	0.1380	0.1240	0.1295	0.1292	0.1354	0.1375
	SCM [31]	0.3052	0.2882	0.2935	0.2946	0.2926	0.4146	0.4186	0.4352	0.4288	0.4177
Deep	CMHH [26]	0.6123	0.5333	0.5354	0.5065	0.4931	0.7133	0.6390	0.6409	0.6089	0.5992
	DCMH [21]	0.5600	0.5593	0.5657	0.5749	0.5750	0.6950	0.6998	0.6829	0.6533	0.6398
	SIDHCNN [5]	0.6010	0.5969	0.5838	0.5872	0.5418	0.7033	0.6924	0.7119	0.6875	0.6715
	DSMHN [23]	0.6399	0.6482	0.6243	0.6494	0.6369	0.7938	0.7972	0.7918	0.7975	0.7801
	MsEspH [17]	0.6681	0.6691	0.6687	0.6673	0.6492	0.7890	0.7793	0.8024	0.7834	0.7946
	FSISR [6]	0.7151	0.7253	0.6843	0.7127	0.6948	0.8212	0.8204	0.8058	0.8313	0.8164
	MambaHash [67]	0.6457	0.6573	0.6347	0.6246	0.6365	0.7147	0.7268	0.7182	0.7293	0.7128
	CCR-Net [53]	0.6627	0.6691	0.6519	0.6461	0.6390	0.7319	0.7524	0.7735	0.7823	0.7847
	ViT-Large [68]	0.7105	0.7231	0.6834	0.7093	0.6897	0.8227	0.8171	0.7947	0.8164	0.8059
	FG-MD (Ours)	**0.7301**	**0.7415**	**0.7103**	**0.7321**	**0.7204**	**0.8545**	**0.8654**	**0.8491**	**0.8513**	**0.8427**

TABLE II

CLASS-WISE MAP COMPARISON BETWEEN THE BASELINE FSISR AND FG-MD (OURS). BOLD INDICATES SUPERIOR PERFORMANCE

Task	**Class Label**	**16 bits**		**24 bits**		**32 bits**		**48 bits**		**64 bits**	
		FSISR	**Ours**	FSISR	**Ours**	FSISR	**Ours**	FSISR	**Ours**	FSISR	**Ours**
MS ↓ PAN	Destroyers	0.8291	**0.8430**	0.7293	**0.7457**	0.7674	**0.7826**	0.7691	**0.8462**	0.7603	**0.7692**
	Littoral Combat Ships	**0.7737**	0.7705	0.7402	**0.8333**	0.7121	**0.7298**	0.7764	**0.7935**	0.7766	**0.8056**
	Combat Boats	0.4832	**0.6687**	0.6055	**0.6111**	0.5582	**0.6816**	0.6725	**0.6778**	0.5705	**0.7276**
	Bulk Carriers	0.6553	**0.7361**	0.6307	**0.7229**	0.6066	**0.6418**	0.6350	**0.6927**	**0.6357**	0.6101
	Container Ships	0.7128	**0.7424**	**0.7677**	0.7649	0.6828	**0.6837**	0.6937	**0.7712**	**0.6812**	0.6810
	Oil Tankers	0.6428	**0.7107**	0.7118	**0.7822**	0.7195	**0.8205**	**0.7627**	0.7591	0.6452	**0.8073**
	All	0.7151	**0.7301**	0.7253	**0.7415**	0.6843	**0.7103**	0.7127	**0.7321**	0.6948	**0.7204**
PAN ↓ MS	Destroyers	0.6861	**0.8231**	0.8308	**0.8564**	0.7603	**0.7692**	0.8462	**0.9231**	0.8452	**0.8462**
	Littoral Combat Ships	**0.7684**	0.7660	0.8311	**0.8333**	0.7587	**0.8333**	0.8308	**0.8390**	0.8310	**0.9167**
	Combat Boats	0.6527	**0.7813**	0.6676	**0.6812**	0.6053	**0.7422**	0.6551	**0.7000**	0.5083	**0.6483**
	Bulk Carriers	0.8113	**0.8326**	0.7327	**0.8609**	**0.8369**	0.8323	0.7520	**0.8457**	0.7824	**0.8528**
	Container Ships	0.8053	**0.8545**	0.8215	**0.8232**	0.8202	**0.8767**	0.8599	**0.8610**	0.8623	**0.8639**
	Oil Tankers	0.8895	**0.9045**	0.8843	**0.9090**	0.8872	**0.9204**	0.9124	**0.9191**	**0.8809**	0.8696
	All	0.8212	**0.8545**	0.8204	**0.8654**	0.8058	**0.8491**	0.8313	**0.8513**	0.8164	**0.8427**

TABLE III

COMPARISON OF MAP RESULTS FOR DIFFERENT BITS ON THE VAIS DATASET

Architecture	Method	IR → VIS				VIS → IR			
		16 bits	24 bits	48 bits	64 bits	16 bits	24 bits	48 bits	64 bits
Shallow	CMFH [20]	0.2424	0.2310	0.2445	0.2352	0.3879	0.4007	0.4444	0.4347
	IMH [33]	0.3282	0.3199	0.3091	0.2936	0.3967	0.3831	0.4029	0.3932
	LSSH [32]	0.2534	0.2975	0.3187	0.3267	0.2916	0.3243	0.3502	0.3710
	SCM [31]	0.5393	0.5495	0.5497	0.5497	0.7378	0.7207	0.7207	0.6997
Deep	DSMHN [23]	0.6733	0.5957	0.6396	0.6240	0.6898	0.5969	0.6453	0.6662
	MsEspH [17]	0.7065	0.7086	0.7101	0.7062	0.7624	0.7242	0.7496	0.7338
	FSISR [6]	0.7333	0.7016	0.7134	0.7271	0.7241	0.6891	0.7333	0.7364
	FG-MD (Ours)	**0.7361**	**0.7230**	**0.7497**	**0.7473**	**0.7696**	**0.7801**	**0.7640**	**0.7521**

bit lengths. Specifically, in the 16-bit VHR → SAR task, the proposed method reaches an mAP of 0.9267, surpassing the state-of-the-art FSISR (0.9201). This robust performance in a heterogeneous cross-domain scenario demonstrates that the frequency-guided strategy effectively captures modality-invariant structural features in cases where spatial-domain methods typically underperform.

2) Qualitative Visualization: To intuitively assess the discriminative capability and the underlying alignment mechanism of FG-MD, we provide comprehensive visualizations from two perspectives: instance-level retrieval results and distribution-level feature spaces.

a) Retrieval results: We first visualize the top-10 retrieval results for representative queries in Fig. 5.

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TABLE IV
COMPARISON OF MAP RESULTS FOR DIFFERENT BITS ON THE CBRSIR_VS DATASET

Architecture	Method	VHR → SAR				SAR → VHR			
		16 bits	24 bits	32 bits	48 bits	16 bits	24 bits	32 bits	48 bits
Shallow	IMH [33]	0.1193	0.1478	0.1588	0.1600	0.1324	0.1728	0.1851	0.1929
	LSSH [32]	0.2416	0.2563	0.2648	0.2660	0.2224	0.2466	0.2532	0.2732
	CMFH [20]	0.2657	0.2461	0.2944	0.3096	0.2571	0.2665	0.3016	0.3226
	SCM [31]	0.4735	0.4958	0.5163	0.5107	0.4765	0.4813	0.4923	0.5042
Deep	CMHH [26]	0.7432	0.7486	0.7532	0.7557	0.7734	0.7732	0.7847	0.7918
	DCMH [21]	0.7631	0.7917	0.7947	0.7920	0.7362	0.7434	0.7495	0.7563
	SIDHCN [5]	0.7963	0.8119	0.8221	0.8396	0.7973	0.8035	0.8044	0.8051
	DSMHN [23]	0.8857	0.8872	0.8906	0.8966	0.7974	0.8037	0.8042	0.8058
	MsEspH [17]	0.9145	0.9314	0.9324	0.9398	0.8333	0.8344	0.8442	0.8458
	FSISR [6]	0.9201	0.9311	0.9278	0.9451	0.8404	0.8411	0.8421	0.8561
	FG-MD (Ours)	**0.9267**	**0.9459**	**0.9402**	**0.9461**	**0.8670**	**0.8541**	**0.8625**	**0.8597**

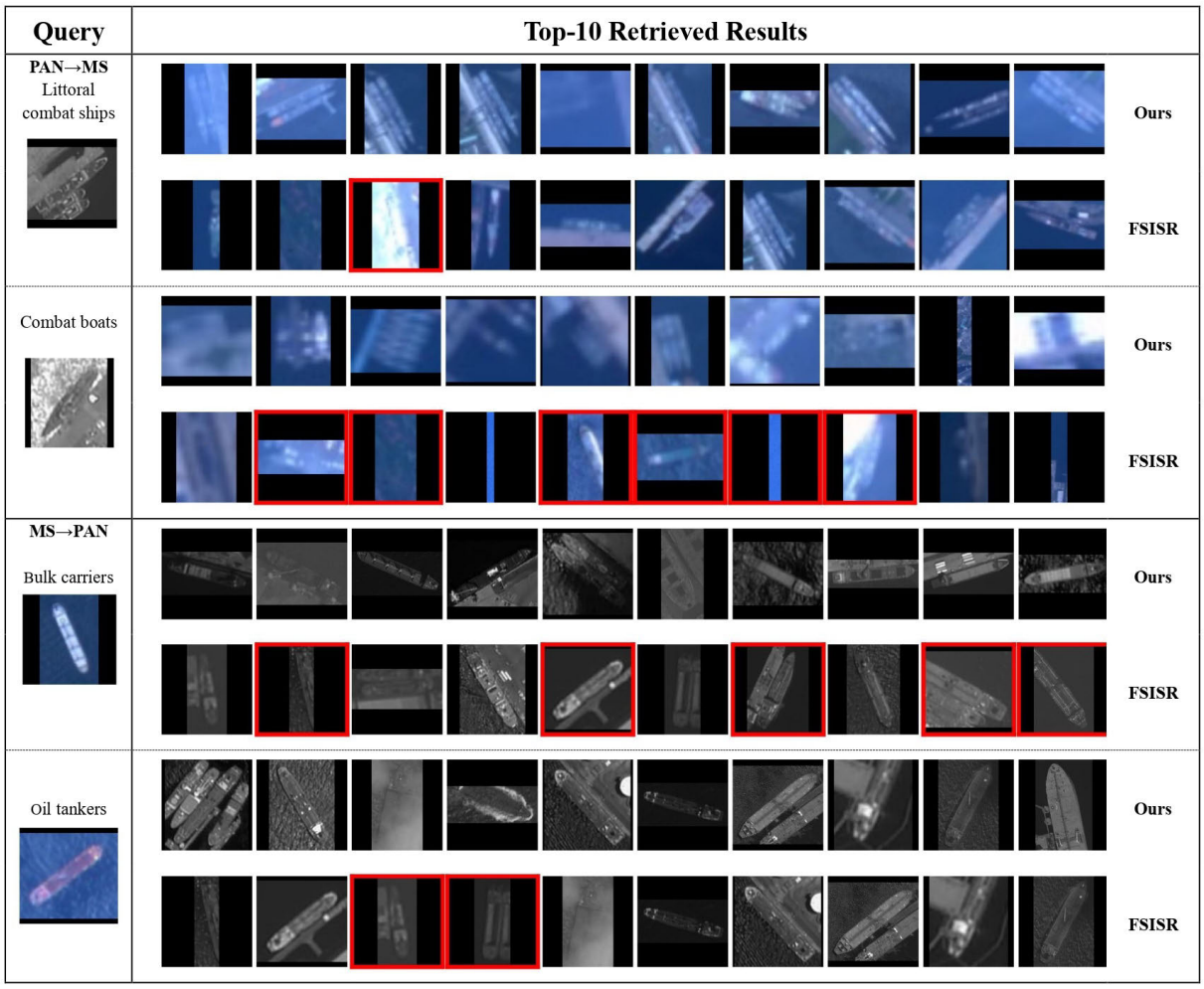


Fig. 5. Qualitative comparison of top-10 retrieval results. Visualization of top-10 retrieval results on the MRSSID dataset. The left column shows the query images, followed by the retrieved ranking list. Red boxes indicate incorrect retrieval results.

Robustness to Background Clutter: As shown in the second row (Query: Combat Boats in PAN → MS task), the query image is characterized by low contrast and significant sea clutter. The baseline FSISR fails to distinguish the target from the background, retrieving multiple irrelevant images that merely share similar background textures (highlighted in

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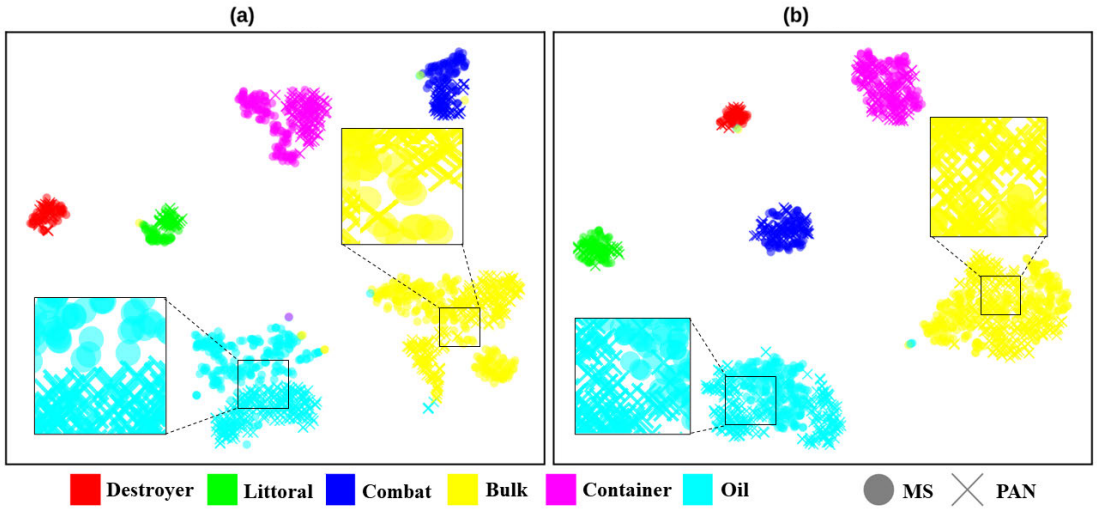


Fig. 6. Visualization of the feature spaces learned by (a) FSISR and (b) our FG-MD on the MRSSID dataset using t-SNE. Different colors represent different ship categories, while circles (•) and crosses (×) denote MS and PAN modalities, respectively. In (a), samples from different modalities within the same class form disjoint subclusters, indicating a significant modality gap. In (b), FG-MD achieves compact intraclass fusion, demonstrating superior cross-modal alignment.

red boxes). In contrast, thanks to the proposed ADF, FG-MD successfully suppresses high-frequency noise and focuses on the ship structure, achieving 100% precision in the top-10 results.

Fine-Grained Semantic Discrimination: In the fourth row (Query: Oil Tankers in MS → PAN task), the visual appearance of oil tankers is highly similar to that of bulk carriers (e.g., flat deck structures). Consequently, FSISR retrieves incorrect images belonging to other categories. However, FG-MD accurately identifies the target category and retrieves semantically consistent images.

b) Feature space analysis: To further uncover the mechanism behind this performance superiority, we visualize the t-SNE feature distributions of the learned hash codes in Fig. 6(b). As shown in Fig. 6(a), although FSISR forms discriminative clusters, it exhibits a VIS modality gap. For instance, the distributions of MS (circles) and PAN (crosses) samples for the Combat (blue) and Container (magenta) classes appear as disjoint subclusters, indicating imperfect alignment. In stark contrast, as illustrated in Fig. 6(b), our FG-MD framework generates highly compact clusters where heterogeneous features are inextricably mixed. The significant overlap between MS and PAN samples within each category confirms that our symmetric mutual distillation strategy effectively eliminates modality-specific variations, learning a unified and robust semantic manifold. This feature-level alignment directly explains the superior retrieval accuracy observed in Fig. 5 and Table I.

C. Efficiency Analysis

The practical efficiency and deployment potential of the proposed FG-MD are evaluated through a comprehensive analysis of computational complexity, memory footprint, and scalability. As demonstrated in Table V, prevailing dual-stream architectures, such as MsEspH and FSISR, exhibit significant redundancy, with parameter counts exceeding 118 M and VRAM consumption reaching approximately 240 MB. Furthermore, although the massive ViT-Large model offers high theoretical capacity, its overwhelming computational load of 155.73 GFLOPs triggers out-of-memory (OOM) failures on the resource-constrained Jetson NX platform. In stark contrast, FG-MD achieves a marked reduction in model complexity, maintaining only 22.05-M parameters and a storage footprint of 84.11 MB. This represents a more than 5.5× reduction in parameters and a 4.3× decrease in inference memory usage (55.20 MB) compared with conventional baselines. Beyond static complexity, the dynamic operational efficiency of FG-MD is validated through edge-side responsiveness and resolution scalability. On the NVIDIA Jetson NX platform, FG-MD attains a real-time inference speed of 48.60 FPS, which comfortably exceeds the 30 FPS threshold required for practical remote sensing applications, whereas compared models either struggle with low throughput or fail to initialize.

Moreover, as reported in Table VI, FG-MD maintains superior computational scalability across varying input resolutions. For instance, at a resolution of 1024^2, it consumes only 7.452 GFLOPs, which is less than 50% of the 15.642 GFLOPs required by FSISR. Combined with a state-of-the-art retrieval accuracy of 0.8654 mAP, these results underscore the substantial advantages of FG-MD for real-time cross-modal retrieval in resource-constrained edge-oriented scenarios.

D. Ablation Studies

To evaluate the effectiveness of the proposed FG-MD framework, we conducted ablation experiments on the MRSSID dataset. The quantitative results are categorized into three targeted substudies focusing on frequency guidance, architectural selection, and mutual distillation strategies. Qualitative visualizations, illustrating spatial attention and feature correlation, are presented in Figs. 7 and 8, respectively.

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TABLE V

COMPREHENSIVE QUANTITATIVE ANALYSIS OF MODEL COMPLEXITY AND EFFICIENCY. RUNTIME AND FPS ARE MEASURED ON THE NVIDIA JETSON NX PLATFORM AT 256 × 256 RESOLUTION

Method	Params (M) ↓	Size (MB) ↓	VRAM (MB) ↓	FLOPs (G) ↓	Runtime (ms) ↓ (Jetson)	Speed (FPS) ↑			mAP ↑
						CPU	GPU	Jetson	
MsEspH [17]	118.33	475.24	243.56	1.127	60.61	44.89	151.60	16.51	0.7793
FSISR [6]	122.10	465.78	239.23	0.958	56.18	46.47	**155.85**	17.83	0.8204
ViT-Large [68]	606.80	2314.90	2362.66	155.73	-	-	12.50	-	0.8171
FG-MD (Ours)	**22.05**	**84.11**	**55.20**	**0.473**	**20.58**	**78.08**	128.07	**48.60**	**0.8654**

† Note: "-" indicates that the dual-stream ViT-Large model is inoperable due to excessive computational overhead (155.73G FLOPs), causing Out-of-Memory (OOM) or timeout failures on both the CPU and the Jetson NX platform.

TABLE VI

SCALABILITY ANALYSIS ACROSS DIFFERENT INPUT RESOLUTIONS

Metric	Method	128^2	256^2	512^2	1024^2
FLOPs (G) ↓	FSISR	0.260	0.958	3.854	15.642
	FG-MD (Ours)	**0.124**	**0.473**	**1.869**	**7.452**
VRAM (MB) ↓	FSISR	214.90	239.23	259.16	343.87
	FG-MD (Ours)	**53.10**	**55.20**	**74.67**	**129.67**

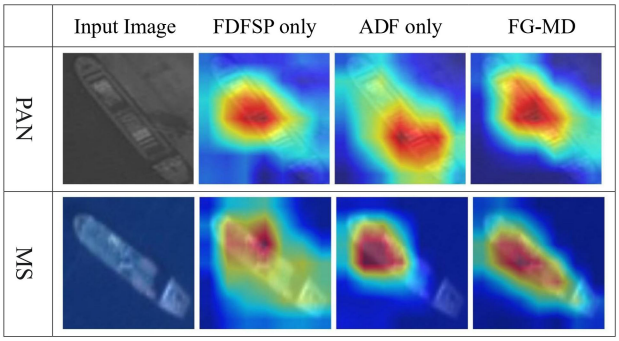


Fig. 7. Visual comparison of feature activation maps (Grad-CAM) for the ablation study. From left to right: the original input image, the variant utilizing FDFSP only, the variant utilizing ADF only, and the proposed FG-MD framework. The top row displays the PAN view, while the bottom row displays the MS view. Warmer colors indicate higher attention activation.

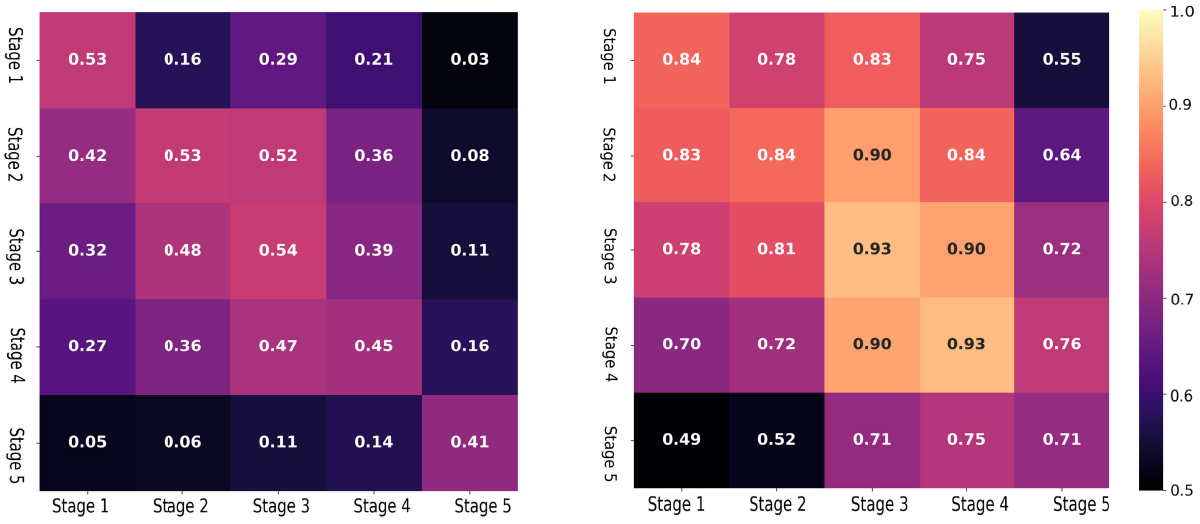
1) Quantitative Analysis of Component Efficacy: The contribution of each module to the overall retrieval performance is analyzed first, ranging from single-module additions to full system integration.

a) Necessity and mechanism of frequency guidance: The contribution of the FDFSP module is systematically analyzed in Table VII. In cross-modal retrieval, the intrinsic modality gap often leads to severe feature discrepancies, as heterogeneous sensors capture distinct physical properties that are difficult to align. This limitation is further exacerbated in lightweight architectures, where standard backbones often suffer from the smoothing effect of conventional convolutions, which inadvertently discard fine-grained structural cues essential for heterogeneous alignment. By incorporating FDFSP (ID 2), a significant boost in mAP is observed, including a gain of up to 10.17% in specific tasks. This justifies that frequency-domain modulation, by leveraging the spectral properties of the FFT, effectively enhances the detailed structural information (e.g., edges and textures) typically lost during feature extraction in multimodal scenarios. To further scrutinize the mechanism, we introduced a variant that directly concatenates raw features with FDFSP outputs (ID 3). While this simple fusion provides supplementary information, it still underperforms compared to the full FG-MD (ID 5). This disparity confirms that the performance gains are derived from the intrinsic spectral refinement and dual-path modulation rather than a mere increase in feature dimensionality. Finally, the consistent performance drop in the ablated variant without FDFSP (ID 4) further confirms the module's critical role in preserving edge sharpness and textural completeness for robust cross-modal matching.

b) Justification of ADF mechanisms: Table VIII validates the efficacy of ADF as a vital architectural component. In cross-modal remote sensing, sensor discrepancies often introduce task-irrelevant background clutter, a challenge exacerbated by the limited receptive fields of lightweight encoders. While generic backbones struggle to isolate salient targets from environmental noise, ADF (ID 2) recalibrates features via global context. To verify its mechanism, we compared the full framework against a direct feature concatenation variant (ID 3). Although concatenation captures additional semantics, it underperforms compared to our adaptive filtering (ID 5). This disparity confirms that ADF functions as a specialized semantic gate to suppress heterogeneous background interference. Ultimately, the superior performance over the variant without ADF (ID 4) proves its essential role in mitigating information asymmetry and bridging the MS-PAN modality gap.

c) Efficacy of mutual distillation strategy: The contributions of the DAMD strategy and the auxiliary losses, $\mathcal{L}_{sem}$ and $\mathcal{L}_{spa}$, are detailed in Table IX. Applying symmetric collaborative learning to the baseline (ID 2) significantly enhances performance by bridging the heterogeneity gap. To further isolate the impact of the distillation framework, we evaluated a variant containing all structural modules (FDFSP and ADF) but excluding both distillation losses (ID 5). The substantial performance degradation observed in ID 5 compared to the full model (ID 6) confirms that structural refinement alone is insufficient without effective feature synchronization. Furthermore, the individual absence of $\mathcal{L}_{sem}$ (ID 3) or $\mathcal{L}_{spa}$ (ID 4) results in a noticeable decline, demonstrating that the DAMD strategy effectively transfers knowledge across the

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(a) Ours (w/o $\mathcal{L}_{dist}$) (b) Ours (Full Model)

Fig. 8. Visualization of layer-wise feature alignment (CKA). (a) Without the distillation loss, deep semantic features show weak correlation. (b) Our FG-MD framework achieves strong diagonal alignment, verifying effective semantic synchronization.

TABLE VII
ABLATION STUDY ON THE NECESSITY AND MECHANISM OF FREQUENCY GUIDANCE (FDFSP)

ID	Variant	Task: MS → PAN (mAP)					Task: PAN → MS (mAP)				
		16 bits	24 bits	32 bits	48 bits	64 bits	16 bits	24 bits	32 bits	48 bits	64 bits
1	Baseline	0.6329	0.6488	0.6328	0.6851	0.6426	0.7413	0.7502	0.7717	0.7637	0.7510
2	Baseline + FDFSP	0.7178	0.7117	0.6726	0.7109	0.6950	0.8346	0.8477	0.8236	0.8347	0.8302
3	Baseline + Concat(Raw, FDFSP)	0.7201	0.7254	0.6912	0.7188	0.7045	0.8412	0.8503	0.8315	0.8402	0.8351
4	FG-MD w/o FDFSP	0.7256	0.7345	0.7077	0.7267	0.7189	0.8498	0.8568	0.8411	0.8477	0.8401
5	**FG-MD (Full)**	**0.7301**	**0.7415**	**0.7103**	**0.7321**	**0.7204**	**0.8545**	**0.8654**	**0.8491**	**0.8513**	**0.8427**

TABLE VIII
ABLATION STUDY ON THE ARCHITECTURAL CHOICE AND MECHANISM OF ADF

ID	Variant	Task: MS → PAN (mAP)					Task: PAN → MS (mAP)				
		16 bits	24 bits	32 bits	48 bits	64 bits	16 bits	24 bits	32 bits	48 bits	64 bits
1	Baseline	0.6329	0.6488	0.6328	0.6851	0.6426	0.7413	0.7502	0.7717	0.7637	0.7510
2	Baseline + ADF	0.7156	0.7109	0.6827	0.7025	0.6994	0.8258	0.8358	0.8053	0.8241	0.8275
3	Baseline + Concat(Raw, ADF)	0.7188	0.7201	0.6905	0.7134	0.7082	0.8312	0.8425	0.8167	0.8304	0.8321
4	FG-MD w/o ADF	0.7267	0.7394	0.7088	0.7265	0.7200	0.8503	0.8591	0.8472	0.8485	0.8413
5	**FG-MD (Full)**	**0.7301**	**0.7415**	**0.7103**	**0.7321**	**0.7204**	**0.8545**	**0.8654**	**0.8491**	**0.8513**	**0.8427**

spatial-semantic dual-dimensional feature space to ensure high-level consistency.

In summary, the integration of FDFSP, ADF, and DAMD constitutes a highly synergistic system. While FDFSP and ADF enhance high-frequency details and suppress clutter, the DAMD strategy ensures effective feature synchronization across MS and PAN modalities. This synergy enables FG-MD to bridge the intrinsic modality gap within lightweight constraints, achieving a state-of-the-art mAP of 0.8654 and demonstrating a superior balance between efficiency and retrieval accuracy.

2) Qualitative Visualization Analysis: To provide an intuitive understanding of the internal mechanisms of FG-MD, we visualize the feature representations from two perspectives: spatial attention via Grad-CAM and feature correlation via CKA heatmaps.

a) Spatial attention analysis (Grad-CAM): We analyze module efficacy via Grad-CAM (Fig. 7). The variant equipped solely with FDFSP (2nd col) highlights sharp boundaries and textures via spectral properties but retains some background noise. Conversely, the ADF-only variant (3rd col) leverages global context to strictly suppress clutter but sacrifices structural sharpness. The full FG-MD framework integrates these strengths, achieving a robust representation where targets remain sharp while the background is clean, effectively coupling fine-grained enhancement with context-aware denoising.

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TABLE IX
EFFECTIVENESS OF MUTUAL DISTILLATION STRATEGY AND AUXILIARY LOSSES

ID	Variant	Task: MS → PAN (mAP)					Task: PAN → MS (mAP)				
		16 bits	24 bits	32 bits	48 bits	64 bits	16 bits	24 bits	32 bits	48 bits	64 bits
1	Baseline	0.6329	0.6488	0.6328	0.6851	0.6426	0.7413	0.7502	0.7717	0.7637	0.7510
2	Baseline + $\mathcal{L}_{sem} + \mathcal{L}_{spa}$	0.7225	0.7285	0.7052	0.7245	0.7153	0.8466	0.8562	0.8395	0.8446	0.8253
3	FG-MD w/o $\mathcal{L}_{sem}$	0.7285	0.7402	0.7095	0.7290	0.7202	0.8530	0.8620	0.8480	0.8500	0.8420
4	FG-MD w/o $\mathcal{L}_{spa}$	0.7272	0.7380	0.7085	0.7275	0.7190	0.8515	0.8605	0.8460	0.8490	0.8410
5	FG-MD w/o $\mathcal{L}_{sem}$ and $\mathcal{L}_{spa}$	0.7207	0.7277	0.7017	0.7224	0.7125	0.8421	0.8495	0.8252	0.8365	0.8348
6	**FG-MD (Full)**	**0.7301**	**0.7415**	**0.7103**	**0.7321**	**0.7204**	**0.8545**	**0.8654**	**0.8491**	**0.8513**	**0.8427**

TABLE X
ADAPTABILITY ANALYSIS ON THE IGS BENCHMARK (RGB → GRAY TASK). MAP@k DENOTES THE MEAN AP EVALUATED ON THE TOP-k RETRIEVED CANDIDATES

Method	Backbone	mAP@1	mAP@3	mAP@5
MsEspH [17]	AlexNet	0.4333	0.4472	0.4768
DSMHN [23]	AlexNet	0.2667	0.3464	0.4050
FSISR [6]	AlexNet	0.4025	0.4513	0.4632
MobileNetV2 [29]	MobileNetV2	0.4375	0.4722	0.4814
FG-MD (Ours)	**MobileNetV2**	**0.4667**	**0.5576**	**0.5684**

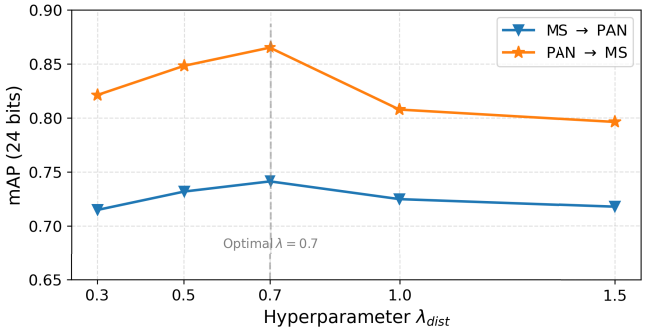


Fig. 9. Sensitivity analysis of the hyperparameter λ_{dist} on both retrieval tasks (24 bits). The performance exhibits an inverted U-shape trend, peaking at $\lambda_{dist} = 0.7$.

b) Feature correlation analysis (CKA): We employ CKA to quantify cross-modal feature similarity (Fig. 8). Without distillation [Fig. 8(a)], the diffuse low-value heatmap indicates a significant "modality gap" with disparate feature extraction. In contrast, FG-MD [Fig. 8(b)] exhibits distinct diagonal alignment, with coefficients exceeding 0.8. This strong correlation validates that symmetric mutual distillation effectively synchronizes semantic trajectories. Notably, high similarity in deep stages confirms the learning of modality-invariant representations, ensuring that dual-stream features are fused in a mutually reinforcing manner.

E. Adaptability Analysis on the IGS Benchmark

To evaluate the adaptability of FG-MD in GPS-denied environments, we conducted experiments on the constructed IGS benchmark (RGB → Gray). All models were retrained on the training set to ensure a fair comparison. The results in Table X reveal that the retrieval methodology, rather than the backbone capacity, is the decisive factor for performance.

1) Limitations of Spatial-Based Methods: Conventional deep hashing methods, such as MsEspH and DSMHN, yield suboptimal performance (e.g., 0.2667 mAP@1). Similarly, the spatial-attention-based FSISR achieves only 0.4025. This common failure stems from their reliance on spatial-domain convolutions, which intrinsically bias the model toward capturing texture and color statistics. In the RGB → Gray cross-modal task, where color information is lost, and texture is degraded, these spatial-based methods fail to extract robust invariant features, leading to poor adaptability.

2) Efficacy of Frequency-Aware Modeling: In contrast, FG-MD significantly outperforms all baselines, achieving an mAP@5 of 0.5684. This superiority is attributed to our FDFSP. Unlike spatial operations, spectral analysis (specifically the phase component) naturally encodes geometric structural information (e.g., layouts of rooms and corridors) which remains consistent across RGB and Gray modalities. By explicitly leveraging frequency-domain priors, FG-MD effectively bypasses the texture/color dependency, demonstrating superior adaptability in geometry-dominant localization scenarios.

F. Parameter Sensitivity Analysis

1) Impact of Distillation Weight: We investigate the sensitivity of the proposed framework to the hyperparameter λ_{dist}, which balances the contribution of the mutual distillation loss in the total objective function [see (12)]. We conducted experiments on the MRSSID dataset (24 bits) by varying λ_{dist} within the range of $\{0.3, 0.5, 0.7, 1.0, 1.5\}$.

As illustrated in Fig. 9, retrieval performance exhibits an inverted U-shaped trend. mAP steadily rises to a peak of 0.8654 as λ_{dist} increases from 0.3 to 0.7, confirming that appropriate distillation facilitates robust knowledge transfer. However, further increasing λ_{dist} to 1.0 and 1.5 causes the performance to drop significantly, suggesting that excessive distillation constraints may dominate the optimization and conflict with the primary hashing objective. Consequently, we empirically set $\lambda_{dist} = 0.7$.

2) Impact of Attention Head Number in FDFSP: We further evaluate the impact of the number of attention heads (N_{head}) in the FDFSP module by varying it among $\{2, 4, 8\}$. The detailed results are listed in Table XI. It can be observed that the

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TABLE XI

PARAMETER SENSITIVITY ANALYSIS OF THE PROPOSED FG-MD FRAMEWORK ON THE MRSSID DATASET (24 BITS). WE INVESTIGATE THE IMPACT OF THE NUMBER OF ATTENTION HEADS (N_{head}) IN FDFSP AND KERNEL SIZE COMBINATIONS IN ADF

Parameter Setting	Retrieval Performance (mAP)	
	MS → PAN	PAN → MS
$N_{head} = 2$	0.7340	0.8584
$N_{head} = 4$	**0.7415**	**0.8654**
$N_{head} = 8$	0.7356	0.8599
$3 \times 3 + 3 \times 3$	0.7120	0.8401
$3 \times 3 + 5 \times 5$	0.7338	0.8559
$3 \times 3 + 7 \times 7$	0.7120	0.8506
$5 \times 5 + 5 \times 5$	0.7160	0.8521
$7 \times 7 + 7 \times 7$	0.7255	0.8561
$5 \times 5 + 7 \times 7$	**0.7415**	**0.8654**

performance is sensitive to the head number. When $N_{head} = 2$, the representation capability is limited, resulting in suboptimal mAP. As N_{head} increases to 4, the performance reaches its peak (0.8654 for PAN → MS), achieving the best balance between feature diversity and channel grouping. However, further increasing N_{head} to 8 leads to a performance drop. This is likely because excessive head splitting fragments the feature channels, making the optimization more difficult. Therefore, we adopt $N_{head} = 4$ as the default setting.

3) Impact of Kernel Size Combinations in ADF: To optimize the receptive field of ADF, we analyzed kernel combinations (k_1, k_2) (Table XI). For homogeneous pairs, performance improves with kernel size, as the baseline $(3, 3)$ lacks the receptive field to model sea clutter, while $(7, 7)$ leverages a broader context for noise suppression. However, simply maximizing size is suboptimal. The heterogeneous $(5, 7)$ combination achieves peak performance, outperforming even $(7, 7)$. This success stems from its asymmetric complementarity: the 7×7 kernel suppresses background noise via global perception, while the 5×5 kernel preserves fine-grained ship structures, mitigating the over-smoothing risks of excessively large kernels. Thus, $(5, 7)$ strikes an optimal balance between clutter suppression and detail preservation.

G. Limitations and Future Work

Despite the demonstrated robustness against typical sensor degradations, the framework exhibits limitations under extreme imaging conditions, as visualized in Fig. 10. Fundamentally, the symmetric mutual distillation strategy relies on the premise that at least one modality retains sufficient semantic cues to guide the other. In instances of severe simultaneous degradation—such as dense cloud cover coupled with low-light environments—both PAN and MS streams suffer critical information loss. Consequently, this may lead to the propagation of noisy supervision signals. Future work will explore the integration of external priors or generative restoration modules to address scenarios where signal integrity in both modalities is severely compromised.

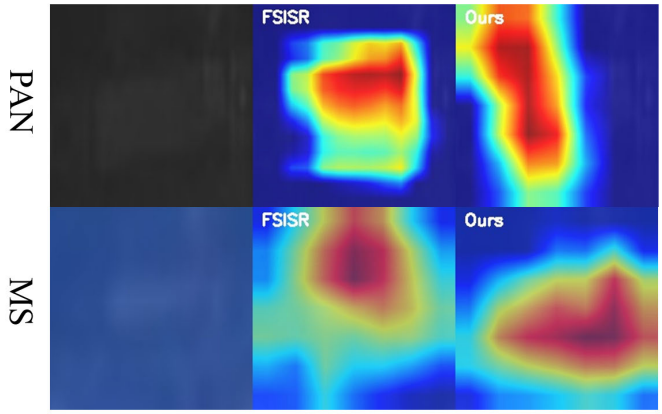


Fig. 10. Visualization of failure cases under extreme imaging conditions. In scenarios where both modalities undergo severe degradation simultaneously (e.g., dense cloud cover coupled with low light), the model struggles to extract effective semantic features due to the absence of valid cross-modal guidance.

V. CONCLUSION

This article presents the FG-MD, a lightweight framework that bridges the cross-modal heterogeneity gap without heavy backbones. By integrating the FDFSP and ADF modules, FG-MD captures fine-grained structural details while suppressing environmental clutter, allowing the symmetric DAMD mechanism to enforce strict semantic consistency. Experiments on the MRSSID dataset show that FG-MD achieves state-of-the-art performance (0.8654 mAP) with 5.4× fewer parameters (~22 M) than mainstream baselines. The IGS benchmark validates its robust generalization in complex, GPS-denied environments. Future work will extend this frequency-aware mechanism to resource-constrained tasks (e.g., UAV tracking) and broader modalities (e.g., SAR and hyperspectral) for all-weather robustness. Furthermore, incorporating domain-invariant frequency modulation and foundation models will enhance cross-domain generalization. Finally, exploring interpretable mechanisms, such as learnable spectral filters, will further elucidate the role of spectral features in cross-modal alignment.

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